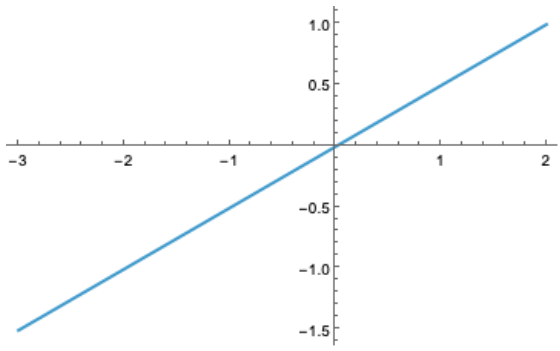
LEIC-T 2025/2026
Aprendizagem - Machine Learning
Homework I
Deadline 1/10/2025 23:00
Submit on Fenix as pdf

I) Correlation (3 pts)

Compute the correlation (Pearson correlation) and **Spearman's rank** for two variables $\mathbf{x}_1$ and $\mathbf{x}_2$=f($\mathbf{x}_1$). Are the values the same or different, please justify. (Indicate all computational steps!) *The functions correspond to activation functions, that will be later introduced when dealing with artificial neural networks.*

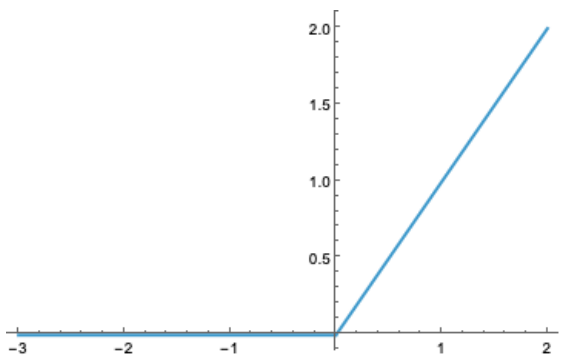
(a) (1 pts)

$\mathbf{x}_1 = (-3, -2, -1, 0, 1, 2)$
for linear function



f($\mathbf{x}_1$)=0.5* $\mathbf{x}_1$
$\mathbf{x}_2$=f($\mathbf{x}_1$).

(b) (1pts)
$\mathbf{x}_1 = (-3, -2, -1, 0, 1, 2)$
for ReLU function



f($\mathbf{x}_1$)=max(0, $\mathbf{x}_1$)
$\mathbf{x}_2$=f($\mathbf{x}_1$).

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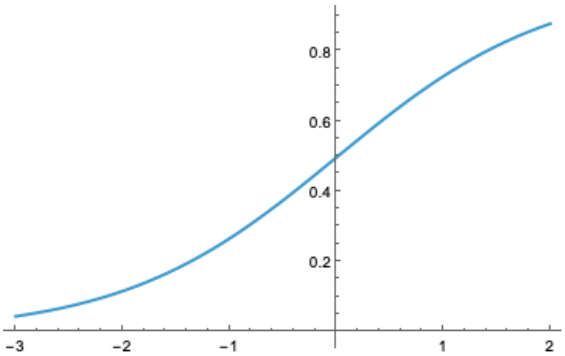
(c) (1pts)
$\mathbf{x}_1 = (-3, -2, -1, 0, 1, 2)$

for sigmoid also called logistic function



$f(\mathbf{x}_1)=\frac{1}{1+e^{-x1}}$ =1/(1+Exp(-$\mathbf{x}_1$)
$\mathbf{x}_2$=f($\mathbf{x}_1$).

II) Decision Trees (4 pts)

Training Set:

Exam	*Weather*	*Summer*	*What to Do?*
No	*Sunny*	*Yes*	*Go to the beach*
Yes	*Sunny*	*Yes*	*Learn*
No	*Rain*	*No*	*Reading*
No	*Claudy*	*No*	*Reading*
Yes	*Claudy*	*Yes*	*Learn*
No	*Rain*	*Yes*	*Reading*
No	*Sunny*	*No*	*Go to the beach*
Yes	*Claudy*	*Yes*	*Learn*

(a) (1.5 pts)
Determine the root of decision tree using the ID3 algorithm with the target "*What to Do?*". Indicate the calculation. (Indicate all computational steps!)

(b) (1.5 pts)
Determine the decision tree using the ID3 algorithm with the target "*What to Do?*"
Draw your decision tree.

(c) (1 pts)
Draw confusion matrix for the test set

Test Set:

Exam	*Weather*	*Summer*	*What to Do?*
No	*Sunny*	*Yes*	*Go to the beach*
Yes	*Sunny*	*Yes*	*Go to the beach*
No	*Rain*	*No*	*Reading*
No	*Claudy*	*No*	*Go to the beach*
Yes	*Claudy*	*Yes*	*Go to the beach*
No	*Rain*	*Yes*	*Learn*
Yes	*Sunny*	*No*	*Go to the beach*
Yes	*Claudy*	*Yes*	*Go to the beach*

III Software Experiments (2pts)

(a) (1pts)
Imagine you are asking ChatGPT a question to generate a python program of the ID3 example of the theoretical lectures. **What would be the question? Write it down.**
Look at the generated code, is the code ok?

Download the jupyter notbook HM1_DT_25.ipynb.
Now, split the data using the command (in the notebook)

```
X_train, X_test, y_train, y_test = train_test_split(X, y, train_size=0.1,stratify=y,random_state=42)
```

And train them using entropy (Shannon's entropy). Note, scikit-learn only supports binary splits and numerical variables. It is not the ID3 algorithm for categorial value :-(.

The splits are based on linear regression which we will introduce later. Also, the algorithm looks for the "best" split (feature + threshold) that maximizes the gain. Sometimes, multiple splits yield the exact same score. In that case, scikit-learn ***randomly*** chooses one of the equally good splits.

```
predictor = tree.DecisionTreeClassifier(criterion='entropy',random_state=42)
```

Now perform the same experiment with the same *train_test_split* and *random*_state but without the command *stratify=y*

(b) (1 pts)
Draw both decision trees, with and without *stratify* (you can copy the drawing into your document). What is the main difference? only one/two sentence/s…

(c) 1 pts)
Why is now the accuracy lower? Please write only one/two sentence/s… Hint, look in the scikit-learn documentation for meaning of the command *stratify=y.*